Adhvik Kanagala

848-256-9450 | adhvik.kanagala@gmail.com | linkedin.com/in/adhvik-kanagala | github.com/addykan

Education

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science, Computer Science, Concentration in Computer Systems — *Aug. 2020 – May 2024*

Experience

Pinecone — June 2024-Present
Software Engineer, Database — *New York City*

- Shipped a job scheduler supporting automated index compaction and resharding, enabling an increase in write traffic from O(1k) requests per second to O(100k) requests per second
- Built freshness layer for querying over freshly written data, reducing time to read consistency from 10s to 500ms
- Wrote a randomized load balancing algorithm for distributing O(10M) variably-sized files over multinenant pool of read nodes – enabled per-customer scaling of serverless compute and a 25% reduction in read pool size
- Shipped enterprise-readiness features such as database-wide customer-managed encryption keys (CMEK)

Vorticity, Inc — Jan 2024 – May 2024
Software Engineer — *Redwood City, CA*

- Designed and built a job scheduler for running compute tasks in self-hosted GPU datacenter
- Parallelized algorithms to take advantage of horizontally scalable GPU availability, speeding up compute jobs 8x

Duolingo — May 2023 – August 2023
Software Engineer Intern, Data Refinery — *Pittsburgh, PA*

- Built an observability service to monitor ETL jobs, eliminating active monitoring work by engineering team
- Built a web UI to visualize dependency graph of datasets used by data science org
- Improved sustainability of data science area's 2023 growth objectives by upgrading tooling

Meta — May 2022 – August 2022
Software Engineer Intern, WhatsApp Business Infra — *Menlo Park, CA*

- Migrated a critical billing service in WhatsApp Business backend to serverless infrastructure, reducing development timelines from weeks to hours
- Eliminated dependency on adjacent chat infra team for deployment/maintenance of business services
- Migrated configuration library in billing service to a new source of truth, enabling quick updates and codification

Projects

c0c | *15-411 Compiler Design* — Spring 2024

- Built a compiler for the C0 language, a Turing-complete subset of C, targeting x86-64
- Implemented lexing, parsing, IR, register allocation, and x86 instruction generation
- Extended support for LLVM to allow ARM compilation

Raft Consensus Algorithm | *15-440 Distributed Systems* — Fall 2022

- Implemented a simulator for the Raft consensus algorithm in Go
- Designed a DFA-style peer model that implements leader election and log replication

Distributed Bitcoin Miner | *15-440 Distributed Systems* — Fall 2022

- Implemented the Live Sequence Protocol, a bespoke network protocol that extends UDP
- Built a server/client model to support automatic packet retries, exponential backoff, and ordered message delivery
- Used LSP to build a mock distributed bitcoin miner with round-robin job scheduling

BustubDB | *15-445 Database Systems* — Spring 2023

- Implemented a buffer pool manager, B+ tree index, SQL query executor and optimizer, and transaction-based concurrency control with multiple isolation levels

Technical Skills

Languages: Rust, Python, Go, C++, C, OCaml, Cuda, Erlang, Hack/PHP, Thrift, x86–64, Standard ML, Typescript
Developer Tools: Kubernetes, Docker, Thrift, Flask, SQL, Unix, React, MongoDB, Flutter
Coursework: Compiler Design, Distributed Systems, Database Systems, Parallel Computer Architecture, Algorithm Design and Analysis, Computer Security, Machine Learning, Writing for the Professions